STAT 812/420

Computational Statistics

1348

$$= 1\times10^3 + 3\times10^2 + 4\times10 + 8\times10^0$$

$\boxed{1}$ $\boxed{2}$ $\boxed{4}$ [illegible] $\boxed{6}$

10^4 10^3 10^2 10^1 10^0

$\boxed{1}$ $\boxed{1}$ $\boxed{1}$ $\boxed{0}$ $\boxed{1}$

$$1\cdot2^5 + 1\cdot2^4 + 1\cdot2^3 + 0\cdot2^1 + 1\cdot2^0$$

000 = 0

001 = 1

010 = 2

011 = 3

100 = 4

Integer

$\boxed{\phi/1}$ $\boxed{0/1}$. - - - - - - $\boxed{0/1}$

32 bits

$$\sum_{i=0}^{p-1} 2^i = 2^p - 1$$

3 bits $\square\ \square\ \square$

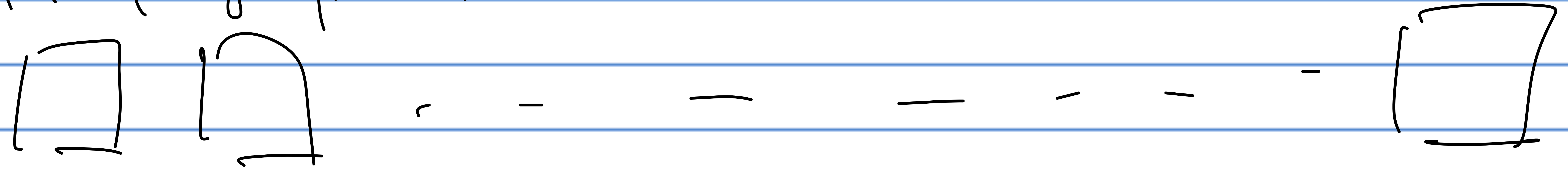
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3

1 2 3 4

↓ ↓ ↓

$1\times10^3 + 2\times10^2 + 3\times10^1 + 4\times10^0$

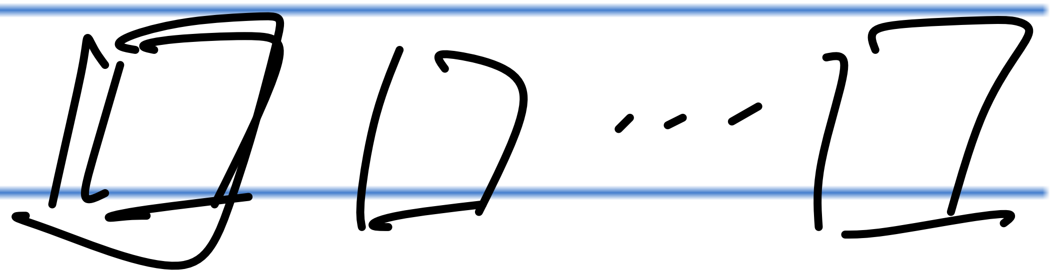
Floating point



Sign

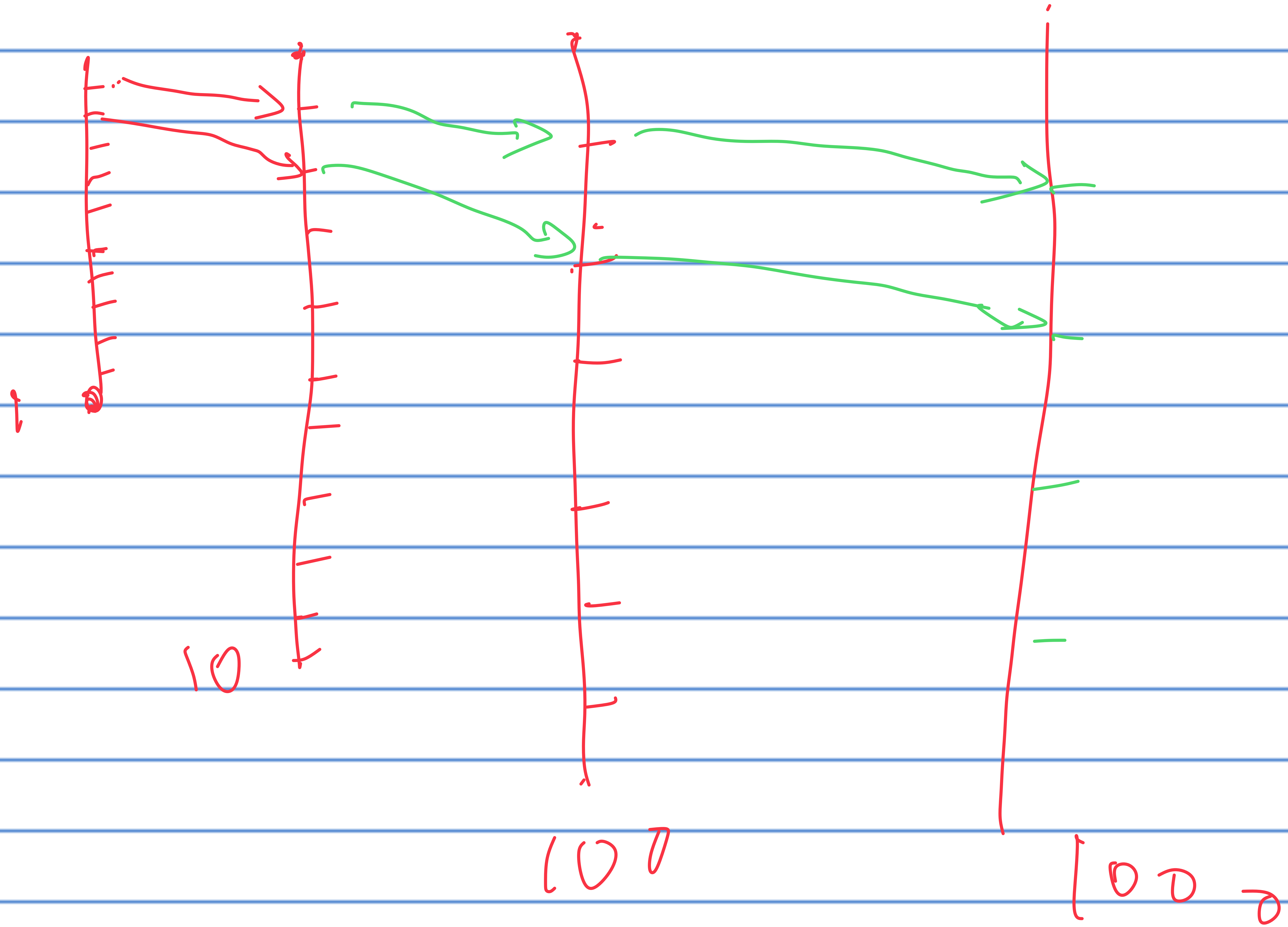
exp.

Mantisa 64

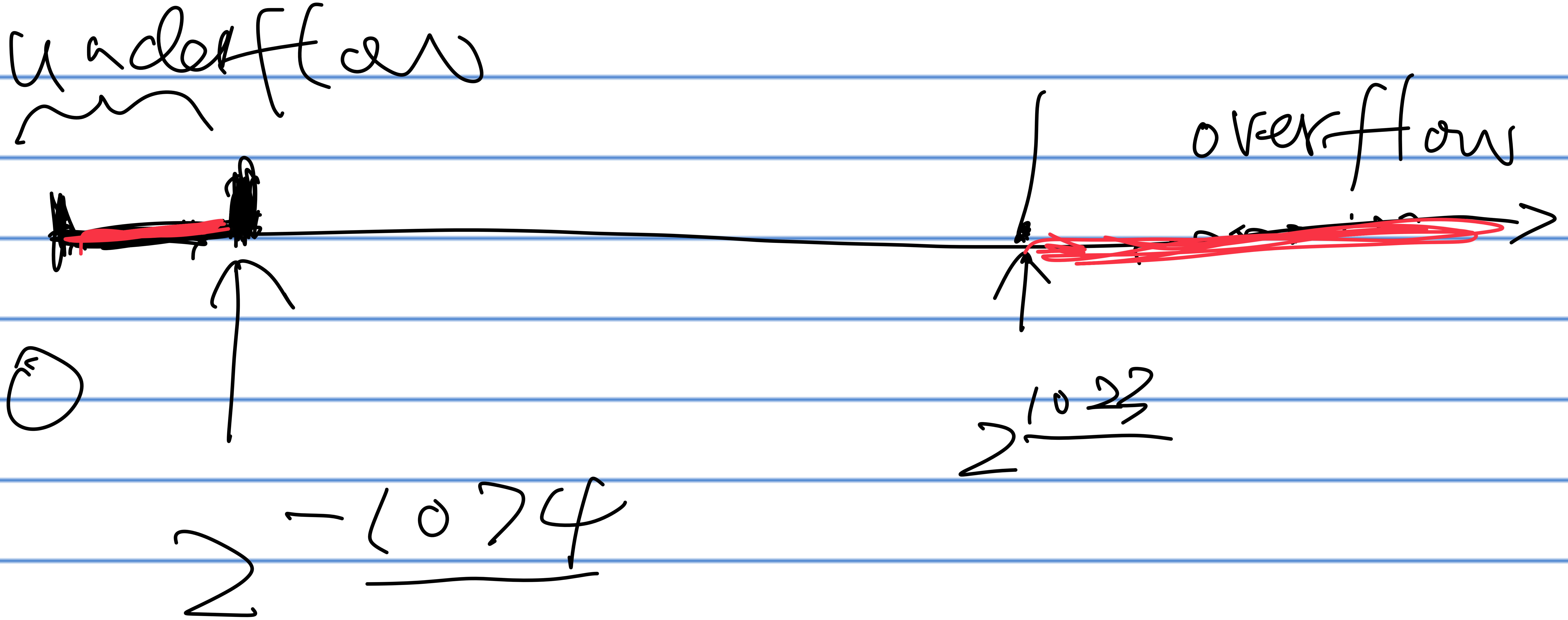


$$(-1)^S \; \underbrace{2.112478}_{} \times 10$$

A Model for Understanding Floating Point Numbers



Overflow & Underflow



$$2^{-2000} \longrightarrow 0$$

Roundoff error

$$\begin{array}{r} \overbrace{\boxed{1}.\boxed{1}\boxed{2}\boxed{1}}^{\text{mantissa}} \\ +\ 0.0\ 0\ 0\ 1 \\ \hline 1.121\boxed{\cancel{1}} \end{array}$$

$$\Downarrow$$

$$1.121$$

$$\begin{array}{r} \boxed{1}.\boxed{1}\boxed{2}\boxed{1}\,\boxed{1} \\ -\ \boxed{1}.\boxed{1}\boxed{2}\boxed{1}\,\boxed{0} \\ \hline 0.000\ 0 \end{array}$$

$$\neq$$

$$0.0001$$

Example

$$1 + 2^{-40}$$

$$1 \boxed{\cancel{+}}\, 2^{-53} \boxed{+} \cdots \boxed{+} 2^{-53}$$

$$\underbrace{\phantom{2^{-53} + \cdots + 2^{-53}}}_{2^{13}}$$

$$= 1 + 2^{-52} \times 2^{12} = 1 + 2^{-40}$$

$$\underbrace{\boxed{1} . \boxed{0} \cdots\cdots \boxed{0}}_{51}$$

$$+\ 0 . 0 \cdots\cdots \boxed{0} \quad 1 = 2^{-52}$$

$$1 . 0 \cdots\cdots 0 \quad \cancel{1}$$

$$2^{-54} + 2^{-54} = 2 \cdot 2^{-54}$$

4 bits in base 10

$$\underline{1.000} + \underbrace{0.0001 + 0.0001}$$

$$= 1.000 \times 10^{0} + 1.000 \cdot 10^{-4} + 1.000 \times 10^{-4}$$

$$= 1.000 + 2.000 \cdot 10^{-4}$$

Example

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!}$$

$$\uparrow$$

u

$x < 0$

$+ \quad - \quad + \quad - \quad + \quad -$

$x > 0$

$$e^{-x} = \frac{1}{e^{x}}$$

$\uparrow$

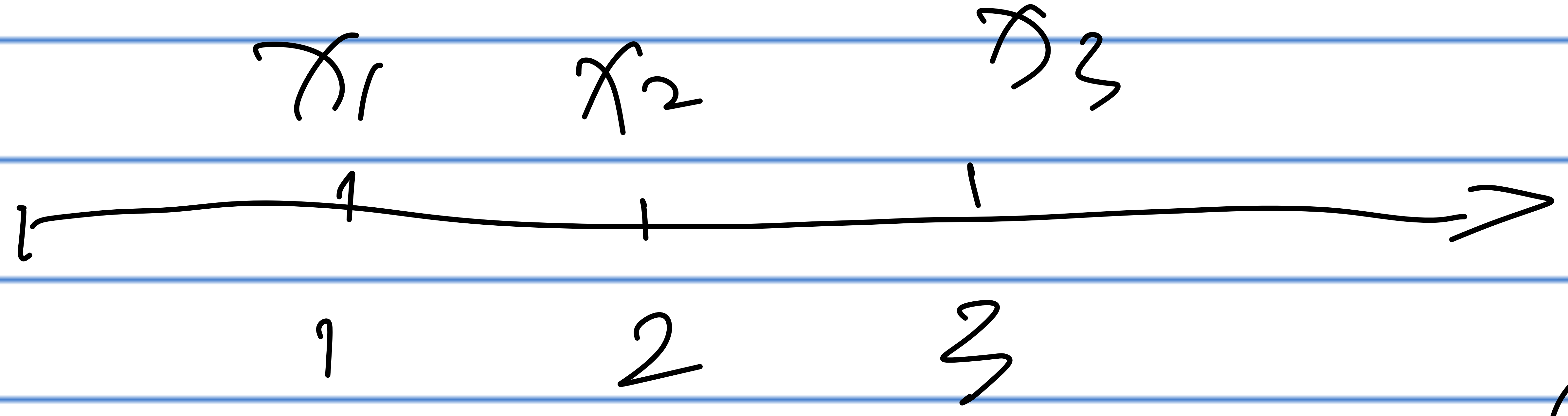
Example

$$\sum_{i=1}^{n} (x_i - \bar{x})^2$$

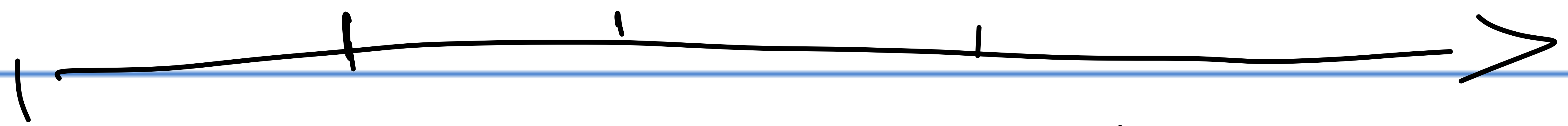
$$= \sum_{i=1}^{n} x_i^2 - n \cdot \bar{x}^2$$

$$= \sum_i x_i^2 - \frac{\left(\sum x_i\right)^2}{n}$$

x_i	x_i^2
x	x
x	x
x	x
x	x
$\sum x_i$	$\sum x_i^2$



$$\sum (x_i - \bar{x})^2 = 2$$



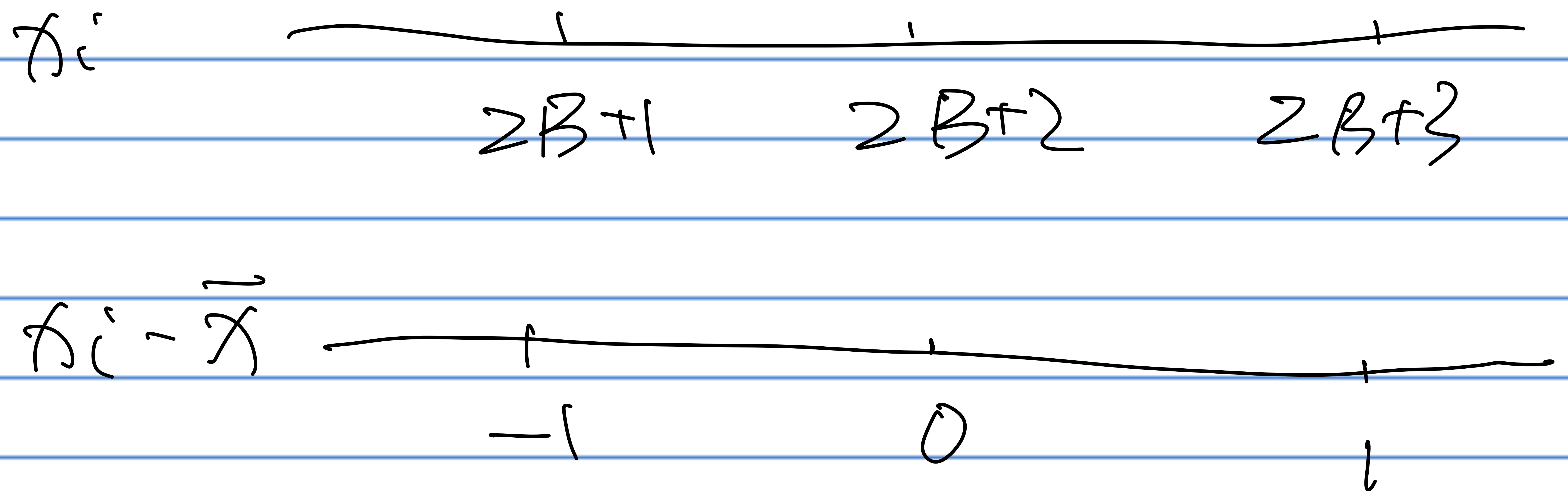
$2M+1$ $2M+2$ $2M+3$

$$\sum (x_i - \bar{x})^2 = 2$$

$\downarrow$ $\downarrow$ $\downarrow$

$(2M+1)^2$. $(2M+3)^2$

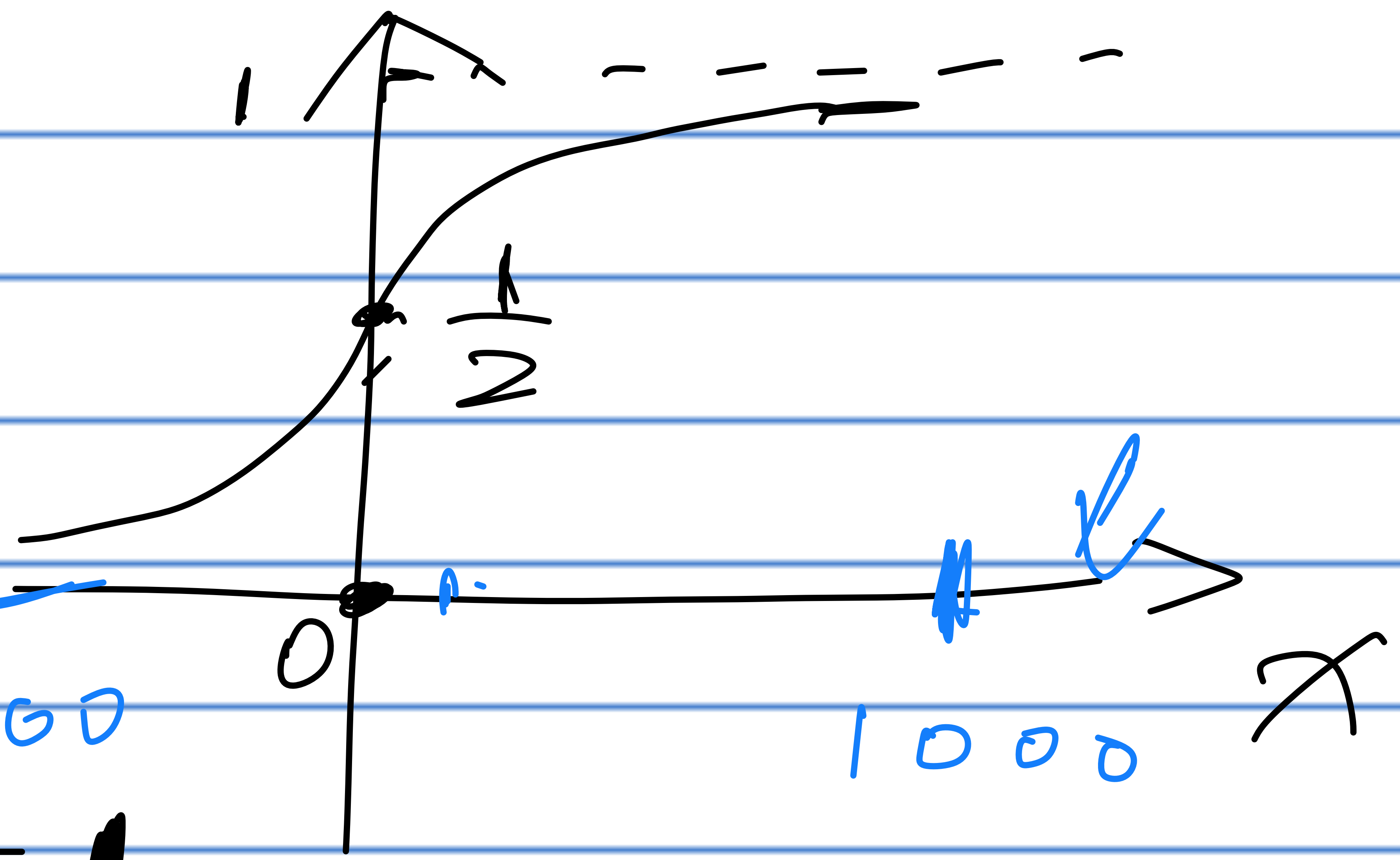
$$\left(\sum x_i\right)^2 = (6M+6)^2 \simeq 12M^2 = 12\cdot 10^{12}$$


x_i
$2B+1$
$2B+2$
$2B+3$
$x_i - \bar{x}$
-1
0
1

Example

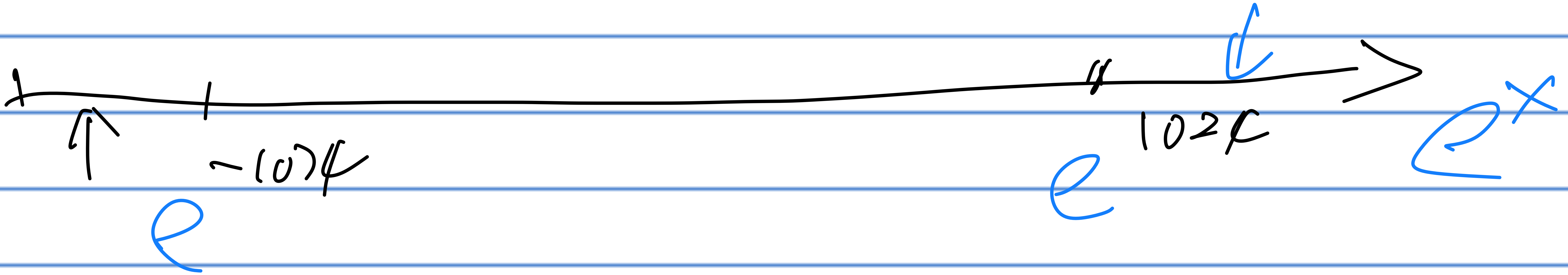
E1 E2

$$\frac{e^x}{1+e^x} = \frac{1}{1+e^{-x}}$$



e^x overflow: $\frac{Inf}{Inf} = NAN$ $\qquad \frac{1}{1+0} = 1$

e^x underflow: $\frac{0}{1+0} = 0$ $\qquad \frac{1}{1+Inf} = 0$



We need to handle underflow & overflow #

$$\prod_{i=1}^{2000} p(x_i) = \left(\frac{1}{2}\right) \cdots \left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{2000} = \frac{1}{2^{2000}}$$

$$(10^{-1})^{1000} = 10^{-1000}$$ underflow

$n!$ $\quad 1000! = 1000 \times 999 \times \cdots \times 1$ overflow

$$\approx 1000^{1000} = 10^{3000}$$ overflow

$$n! \approx n^{n-\frac{1}{2}}$$

Representing # i's logarithm form

x $\log_e(x)$

$$e^{+2000} \longleftrightarrow 2000$$

$$e^{-2000} \longleftrightarrow -2000$$

Given $\ell_x = \log(x)$, return $\log(f(e^{\ell}))$

$$x_1, \ldots\ldots, x_n \xrightarrow{f} y$$

$$\ell_1 \qquad \ell_2 \xrightarrow{f} \ell_y$$

Computing $\log(e^{l_1} \cdot e^{l_2})$

$$\log(x_1 \cdot x_2) = \log x_1 + \log x_2 \quad , \quad \log x_i = l_i$$

$$= l_1 + l_2$$

$$\log\left(\prod_{i=1}^{n} x_i\right) = \sum_{i=1}^{n} \log x_i = \sum_{i=1}^{n} l_i$$

Computing $\log\left(\frac{e^{l_1}}{e^{l_2}}\right)$

$$\log\left(\frac{x_1}{x_2}\right) = \log x_1 - \log x_2 = l_1 - l_2$$

Example: $x_1 = 2^{2000}$, $x_2 = 2^{2001}$. $\log\left(\frac{x_1}{x_2}\right) = \log\left(\frac{1}{2}\right)$

Computing $\log \sum e^{l_i}$:

$\log(\sum x_i) = \log(\sum e^{l_i})$ [log-sum-exp]

$$\begin{array}{ccccc} x_1 & x_2 & \cdots & x_n \\ \Updownarrow & \Updownarrow & & \Updownarrow \\ l_1 = \log x_1 & l_2 = \log x_2 & & l_n = \log x_n \end{array}$$

$$\log\left(\sum_{i=1}^{n} x_i\right) = \log\left(\sum_{i=1}^{n} e^{l_i}\right)$$

$$= \log\left(\sum_{i=1}^{n} e^{l_i - m} \cdot e^{m}\right)$$

$$= \log\left(\sum_{i=1}^{n} e^{l_i - m}\right) + m$$

x_i: $\quad e^{2000} \quad e^{2001} \quad e^{2002}$

$\quad\quad\ \ \parallel \quad\quad\ \ \parallel \quad\quad\ \ \parallel$

$\quad\quad \mathrm{Inf} \quad\ \ \mathrm{Inf} \quad\ \ \mathrm{Inf}$

$$\sum_{i=1}^{3} x_i = \mathrm{Inf}$$

l_i: $\quad 2000 \quad 2001 \quad 2002$

$m = \max(l_i)$

$l_i - m$: $\quad -2 \quad -1 \quad 0$

$e^{l_i - m}$

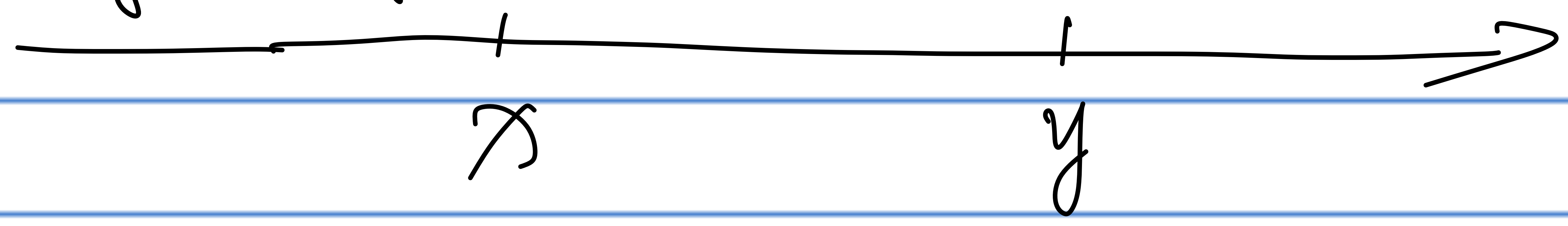
$$e^{-2} + e^{-1} + e^{0}$$

$$\sum_{i=1}^{n} e^{l_i - m} \longrightarrow \log\left(\sum_{i=1}^{n} e^{l_i - m}\right) \longrightarrow \log\left(\sum e^{l_i - m}\right) + m$$

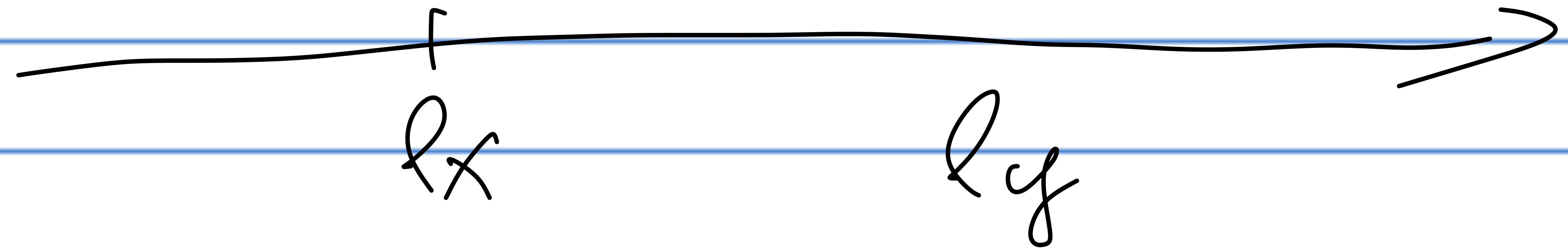
$$\parallel$$

$$\log\left(\sum e^{x_i}\right)$$

Computing $\log(e^{lx} - e^{ly})$



$x > y$



$$\log(x - y) = \log(e^{lx} - e^{ly})$$

$$= \log(e^{lx}(e^{lx-ly} - e^{0}))$$

$$= lx + \log(1 - e^{ly-lx})$$

Question 1:

How to compute $\log\left(\frac{1}{1+e^{-x}}\right)$

for overflow & underflow e^{x}.

Question 2: Softmax

$$\log\left(\frac{e^{x_k}}{\sum_{k=1}^{K} e^{x_k}}\right),$$

for underflow & overflow e^{x_k}